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# AMF-mediated systemic regulation induces Zn-deficiency tolerance through nutrient and metabolic adjustment in pea plants

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## Abstract

Zinc (Zn) deficiency limits plant growth and productivity, yet the role of arbuscular mycorrhizal fungi (AMF) in Zn-deficiency tolerance in pea plants remains unclear. In this study, we investigated the effect of *Rhizophagus irregularis* on pea plants exposed to Zn deficiency in hydroponic conditions. Microscopic observation and qPCR-based quantification of AMF abundance confirmed successful colonization only in AMF-inoculated plants, accompanied by the presence of different AMF structures in the roots. Zn deficiency severely impaired plant health, resulting in reduced biomass, photosynthetic parameters, and Zn status in tissues. Zn deficiency also increased hydrogen peroxide accumulation, electrolyte leakage, and altered mineral homeostasis. However, AMF inoculation significantly alleviated these detrimental effects by improving growth performance, photosynthetic traits, and Zn accumulation under Zn-deficient conditions. AMF further induced the expression of Zn transporter-related genes (*ZIP1* and *ZIP2*), suggesting enhanced Zn acquisition in inoculated plants. In addition, Zn deficiency caused substantial reprogramming of amino acid metabolism, characterized by increased stress-associated amino acids and reductions in several growth-related amino acids, including leucine, isoleucine, glycine, and tyrosine. AMF inoculation restored several of these metabolites under Zn-deficient conditions, indicating metabolic adjustment associated with stress recovery and physiological stabilization. Importantly, split-root analysis showed improved shoot growth, biomass, and chlorophyll status even when AMF colonized only one portion of the root system, providing evidence for systemic regulation. Overall, these findings demonstrate that AMF alleviate Zn deficiency through coordinated regulation of Zn acquisition, metabolic adjustment, and systemic physiological responses, highlighting their potential for improving plant performance under Zn-limited conditions.

**Keywords** Amino acid metabolism, Arbuscular mycorrhizal fungi, Pea, Systemic signaling, Zinc deficiency



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## 1 Introduction

Zinc (Zn) participates in various physiological and biochemical processes that facilitate plant growth and survival. Although needed in small amounts, its deficiency or excess can significantly impact plant metabolism [7]. Zn is necessary for healthy root and shoot development, seed production, and stress tolerance [3]. Zn naturally occurs in soils of terrestrial ecosystems and is indispensable for plant growth due to its involvement in enzyme activation, protein synthesis, membrane integrity, and regulation of gene expression [8, 22].

Plants obtain Zn mainly from the soil solution, where Zn occurs predominantly as free  $\text{Zn}^{2+}$  (and, especially at higher pH, as hydroxo/carbonate/bicarbonate species), which is transported to root surfaces by mass flow and diffusion and then taken up by roots [27]. However, Zn availability is often limited in calcareous and alkaline soils, which are widespread in many agricultural regions, leading to widespread Zn deficiency in crops and consequent yield and nutritional losses [3, 8]. Zn deficiency is recognized as one of the most prevalent micronutrient constraints affecting crop productivity worldwide, particularly in legume-based systems where nutrient demand is high [3, 7]. In plants, Zn deficiency results in reduced chlorophyll synthesis, impaired photosynthesis, stunted growth, and disruption of hormonal balance, ultimately affecting both biomass accumulation and reproductive development [24].

Certain fungi, particularly mycorrhizal fungi, play an important role in increasing Zn uptake in plants. Numerous symbiotic relationships between fungi and plants are essential to the stability of ecosystems, nutrient intake, and plant health [20, 21, 29]. Particularly, arbuscular mycorrhizal fungi (AMF) are essential for enhancing plant Zn nutrition and reducing the negative effects of Zn stress on plants [26, 29]. AMF form mutualistic associations with the roots of most terrestrial plants, extending their extraradical hyphal networks into the soil, thereby increasing the effective root surface area for nutrient acquisition [29]. This hyphal network allows plants to access sparingly available Zn pools beyond the rhizosphere depletion zone, which are otherwise inaccessible to non-mycorrhizal roots [21]. In addition to improving Zn uptake, AMF enhance plant tolerance to both biotic and abiotic stresses by improving soil structure, water relations, and nutrient balance [5, 6]. AMF colonization has also been shown to regulate the expression of Zn transporter genes (e.g., ZIP family transporters), thereby facilitating efficient Zn translocation from fungal hyphae to plant root cortical cells [9, 31]. By facilitating the transfer of Zn from fungal hyphae to plant root cells, these transporters aid plants in overcoming Zn shortage. Furthermore, AMF can alter rhizosphere chemistry by releasing organic acids and chelating compounds that increase Zn solubility and bioavailability [29]. Therefore, AMF are widely recognized as promising biofertilizers for sustainable agriculture, contributing to improved nutrient use efficiency and reduced reliance on chemical fertilizers [6].

Pea (*Pisum sativum* L.) is a major grain legume and an important plant-based protein source for both human and animal nutrition. It is valued for its high protein content, essential amino acids, and its ability to fix atmospheric nitrogen through symbiosis with rhizobia, making it a key component of sustainable cropping systems [12, 30]. However, pea is particularly sensitive to micronutrient deficiencies, including Zn deficiency, which can significantly reduce nodulation efficiency, nitrogen fixation, and overall productivity [24]. Despite the recognized benefits of AMF in improving micronutrient uptake, the

mechanistic understanding of AMF-mediated Zn acquisition and stress mitigation in pea remains limited. Therefore, this study explores the role of AMF in mitigating Zn deficiency in garden pea, with a focus on plant growth, physiological responses, amino acid profiles, and nutrient dynamics.

## 2 Materials and methods

### 2.1 Plant cultivation and AMF treatments

Pea seeds were obtained from the commercial seed company Circle Seed (Dhaka, Bangladesh). The seeds were surface sterilized with 70% ethanol for 30 min, followed by three rinses with distilled water. They were then germinated on tissue paper in a germination tray at ambient temperature. Seeds were allowed to sprout on damp tissue paper, maintained with deionized water, in a dark atmosphere at room temperature for three days. Only healthy, uniform seedlings were selected for transplantation. In hydroponic system, a half-strength basal nutrient solution [18] was used, with the following nutrient concentrations ( $\mu\text{M}$ ):  $\text{KNO}_3$  (16,000),  $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$  (6000),  $\text{NH}_4\text{H}_2\text{PO}_4$  (4000),  $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$  (2000),  $\text{KCl}$  (50),  $\text{H}_3\text{BO}_3$  (25),  $\text{Fe-EDTA}$  (25),  $\text{MnSO}_4 \cdot 7\text{H}_2\text{O}$  (2),  $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$  (0.5),  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  (0.5), and Zn as  $\text{ZnSO}_4$  at two levels, viz.  $0.1 \mu\text{M}$  (Zn-deficient) and  $2.0 \mu\text{M}$  (Zn sufficient/control). The AMF inoculum consisted of 500 spores from *Rhizophagus irregularis* (Organic Gardening, New South Wales, Australia) as previously described with slight modification [13]. There were four treatments: control ( $2.0 \mu\text{M}$   $\text{ZnSO}_4$ ); Zn- ( $0.1 \mu\text{M}$   $\text{ZnSO}_4$ ); Zn-AMF+ ( $0.1 \mu\text{M}$   $\text{ZnSO}_4$  +  $0.2 \text{ g}$  AMF); and AMF+ ( $2.0 \mu\text{M}$   $\text{ZnSO}_4$  +  $0.2 \text{ g}$  AMF). The pH of the medium was adjusted to 6.0 using  $\text{KOH}$  (1 M). Plants were kept in a growth chamber at  $25^\circ\text{C}$  with 60% relative humidity,  $200 \mu\text{mol m}^{-2} \text{ s}^{-1}$  light intensity, and long-day light conditions (14 h light/10 h dark). The experiment was arranged in a completely randomized design (CRD) with three independent biological replicates per treatment. Each independently maintained hydroponic culture served as an experimental unit, and the experimental units were randomly assigned to the four treatments and randomly positioned in the growth chamber. Plants were harvested five weeks after transfer to the hydroponic culture.

### 2.2 Morphological features

Morphological growth parameters, such as shoot height, root length, shoot fresh weight, and root fresh weight were measured on harvested plants grown on solution culture. Roots developed by each plant sample were washed in distilled water and then quickly blotted in tissue paper before taking weight by electronic balance. The fresh weight of the shoot was directly weighed by electronic balance after harvesting. For turgid weight, the fresh shoots were then immersed in deionized water overnight at  $4^\circ\text{C}$  until full hydration was achieved. After surface moisture was carefully removed with tissue paper, the fully hydrated shoots were weighed to obtain the turgid weight. Furthermore, shoot height and root length were measured by cm measuring scale. The roots and leaves of the plants were separated manually and dried in an oven at  $80^\circ\text{C}$  for 3 days before measuring the dry weight.

### 2.3 Determination of chlorophyll score and photosynthetic parameters

The chlorophyll content of young, fully expanded leaves was measured by SPAD (Soil Plant Analysis Development) meter (Minolta, Japan). Furthermore, FluorPen FP 100

(Photon Systems Instruments, Czech Republic) was used to investigate the chlorophyll fluorescence characteristics, such as Fv/Fm (quantum efficiency of photosystem II) and Pi\_ABS (photosynthesis performance index) of young, fully expanded leaves adapted at dark for 1 h.

#### 2.4 Visualization of AMF colonization in the roots

Roots from meristematic zone were cut into 1 cm segments and cleared in 5% KOH at 60 °C for 5 min. After rinsing, samples were acidified in 1% HCl for 5 min, stained with 0.05% trypan blue at 60 °C for 10 min, and blotted to remove excess stain. Roots were then incubated in lactoglycerol for 40 min, mounted on slides with lactoglycerol, and examined under a digital microscope at 40×. These staining and microscopy methods are widely used to identify AMF structures in roots and quantify the extent of mycorrhizal colonization [32].

#### 2.5 Molecular detection and relative abundance of AMF in roots

We assessed AMF colonization in roots collected from the meristematic zone using conventional PCR (polymerase chain reaction). Briefly, DNA was extracted from roots using the CTAB method [10]. Prior to PCR, the DNA was diluted to equal concentration. For molecular confirmation of AMF colonization, genomic DNA was amplified with *R. irregularis* 18S rRNA primers (F: ATCAACTTTTCGATGGTAGGATAGA, R: GAACCCAAACACTTTGGTTTCC) (Badri et al. 2016). PCR products were mixed with loading dye and loaded into gel wells alongside a DNA ladder. The gel run was carried out at 120 V for 30 min. Bands were then visualized under UV and sized by comparison with the DNA ladder.

For determination of relative AMF abundance, DNA-based quantitative real-time PCR (qPCR) was performed using the same *R. irregularis* 18S rRNA primers on a CFX96 Real-Time PCR System (Bio-Rad, USA). The qPCR program consisted of an initial denaturation at 95 °C for 2 min, followed by 39 cycles of 95 °C for 5 s and 55 °C for 30 s, with a final extension at 60 °C for 5 min. Relative AMF abundance was calculated using the  $2^{-\Delta\Delta C_t}$  method, with *PsGAPDH* as the internal reference for normalization. Three independent biological replicates were analyzed, with two technical replicates for each biological replicate.

#### 2.6 Determination of electrolyte leakage in roots

Electrolyte leakage was measured in roots using an electrical conductivity meter to assess membrane integrity. Briefly, root samples from one randomly chosen plant per replicate were cut into 1 cm segments. Samples were then placed in individual stoppered vials containing 20 ml of non-ionized water after three washes with distilled water to remove surface contamination. These samples were incubated at room temperature (25 °C) on a shaker (100 rpm) for 2 h. Electrical conductivity (EC) of bathing solution was read after incubation [23].

#### 2.7 Determination of hydrogen peroxide in roots

The roots were homogenized in 0.1% trichloroacetic acid using mortar and pestle for hydrogen peroxide analysis. The homogenate was centrifuged at 10,000 rpm for 15 min to remove cellular debris. The resulting supernatant was mixed with potassium iodide

(1.0 M) and phosphate buffer (10 mM, pH 7.0) and incubated in darkness for 60 min. Absorbance was measured at 390 nm using a Shimadzu UV-1650PC spectrophotometer [2].

## 2.8 Elemental analysis in root and leaf

Multiple mineral elements were quantified by Inductively Coupled Plasma Mass Spectrometry (ICP-MS; PerkinElmer, NexION 2000, USA) and Zn, Fe, Cu, and Mn were selected for presentation since they exhibited significant treatment-dependent responses. Briefly, nutrient analysis was conducted using root and young leaf tissues. Root samples were carefully excised, rinsed under running tap water, immersed in 0.1 mM  $\text{CaSO}_4$  solution for 10 min, and subsequently washed with deionized water. Young leaf samples were separately rinsed with deionized water. All samples were transferred into labeled envelopes and oven-dried at 75 °C for 3 days. The dried samples were then finely powdered. Approximately 0.5 g of powdered sample was transferred into a clean, sterile 50 mL beaker. Ten milliliters of  $\text{HNO}_3$  followed by 2.5 mL of  $\text{H}_2\text{O}_2$  were added to each sample, which was then kept overnight at room temperature. Three blank samples were prepared simultaneously to account for possible contamination from reagents and instruments. Afterward, sample digestion was performed in a fume hood chamber, beginning at 100 °C, with the temperature gradually increased to 150 °C, and then to 200 °C until the solution was clear. After cooling to room temperature, the digested samples were diluted to a final volume of 50 mL with ultra-pure deionized water using volumetric flasks. All samples were filtered through Whatman filter paper to remove particulates prior to analysis. During analysis, the nebulized sample ( $\sim 0.85 \text{ mL min}^{-1}$ ) entered the core of the argon plasma of the ICP-MS system.

## 2.9 Expression pattern of Zn transporter genes in the roots

We performed real-time qPCR analysis to validate the expression of Zn transporter genes, such as *ZIP1* and *ZIP2* in the roots of pea on the CFX96 Real-time System (Bio-Rad, USA) with gene-specific primers. Briefly, the first-strand cDNA was synthesized from total RNA using the GoScript Reverse Transcription System (Promega, USA). The qPCR cycling conditions were as follows: initial denaturation at 95 °C for 2 min, followed by 42 cycles of 95 °C for 5 s, annealing at 55 °C for 30 s, and a final extension at 60 °C for 5 min. Relative expression levels were calculated using the  $2^{-\Delta\Delta\text{Ct}}$  method, with *PsGAPDH* serving as the internal reference gene. The real-time PCR experiment included three independent biological replicates, with two technical replicates performed for each. The primers used for qPCR experiments were as follows: *PsGAPDH* (fw: GTGGTCTCCACTGACTTTATTGGT, Rv: TTCCTGCCTTGGCATCAAA), *ZIP1* (F: TGAGGGAATCGCCATTGGTC, R: AGAGATCGCGTAGGTCCAGT) and *ZIP2* (F: CCCATAGGTGTCGGGATTGG, R: GGCCCATAGAGATCGCGTAG).

## 2.10 Amino acid profiling in pea roots and leaves

The amino acid composition of pea root and leaf tissues was determined using the chromatographic method [25]. Briefly, approximately 0.40 g of each dry powdered sample was treated with 12 mL of 7.0 N HCl, 36 mL of deionized water, and 0.1 mg phenol for acid digestion, followed by thorough homogenization in sealed glass tubes and incubation at 102 °C for 24 h for hydrolysis. Following hydrolysis, the samples were filtered

and diluted to 100 mL with deionized water. The diluted solutions were subsequently filtered through a 0.45 µm membrane filter prior to amino acid determination. Amino acid analysis was performed using an amino acid analyzer (Model: S433-D, Sykam AAA, Germany). The amino acid analyzer was equipped with an autoinjector and a capillary column (150 mm × 4.26 mm). The column temperature was maintained at 57 °C, with a flow rate of 0.45 mL min<sup>-1</sup> at a pressure of 43 bar. The total run time for the analysis was 45 min, with amino acid detection performed at wavelengths of 440 nm and 570 nm using a Dual-Channel Photometer. The amino acid content was quantified by comparing the sample peak areas to those of amino acid standards, based on their respective retention times.

### 2.11 Determination of symbiotic signalling using split root assay

A split root system is a setup where a plant's root system is divided into two or more separate compartments separated by a sealed plastic barrier, allowing different treatments to be applied to different root zones of the same plant. The plants were cultivated for one week in a sterile basal nutrient solution [18] before conducting the split-root assay in treatment pots. The taproot of each seedling was cut in the middle, and the lateral root system was evenly divided between two compartments within a single pot filled with different solutions. The compartments were separated by a solid plastic barrier. Each split-root system was subjected to the same growth conditions, including the presence or absence of Zn deficiency. In one compartment of each split-root system, AMF was inoculated, while the other compartment remained uninoculated, based on the treatment. The split-root experiment was conducted using a completely randomized design with three independent biological replicates per treatment combination. Plants were cultivated for an additional four weeks before data collection. The split-root assay was performed under six conditions: control/control, Zn-/Zn-, Zn-/Zn-AMF+, Zn-AMF+/Zn- AMF+, control/AMF+, and AMF+/AMF+.

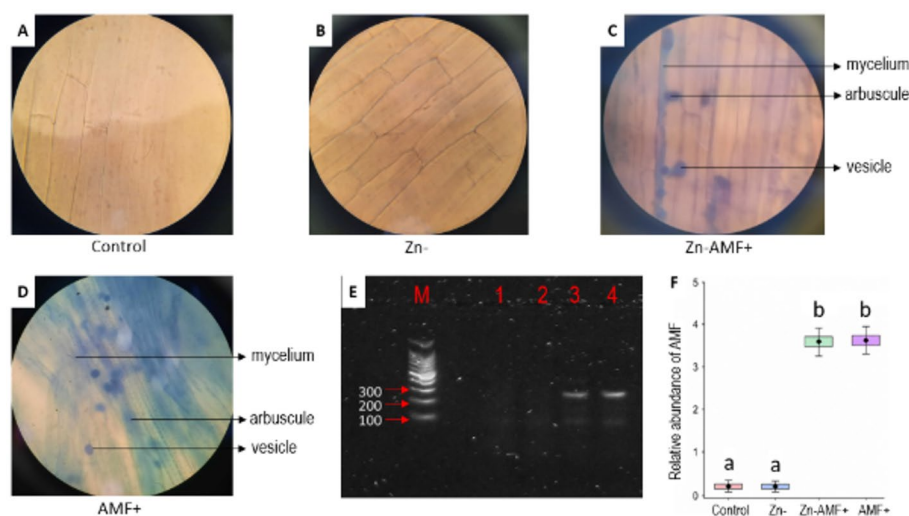
### 2.12 Statistical analysis

All experimental data were analysed using a completely randomized design (CRD), with three independent biological replicates per treatment. The statistical significance of each variable was evaluated by analysis of variance (ANOVA), where appropriate. Where significant differences were detected by ANOVA ( $P < 0.05$ ), Duncan's multiple range test was employed as a post-hoc analysis to identify specific group differences. All statistical analyses were conducted using SPSS Statistics 20.0 (IBM Corporation, USA). Graphical representations were generated using the R package ggplot2.

## 3 Results

### 3.1 AMF colonization and abundance in the roots

Microscopic observation of stained roots showed no visible AMF colonization in the control and Zn-deficient plants without AMF inoculation (Fig. 1A, B). In contrast, roots inoculated with AMF under Zn-deficient and Zn-sufficient conditions showed clear fungal structures, including mycelia, arbuscules, and vesicles (Fig. 1C, D). PCR amplification further confirmed AMF colonization, with visible bands detected in AMF-inoculated samples, particularly in lanes 3 and 4, while no clear amplification was observed in the non-inoculated samples (Fig. 1E). Consistent with the microscopic and molecular



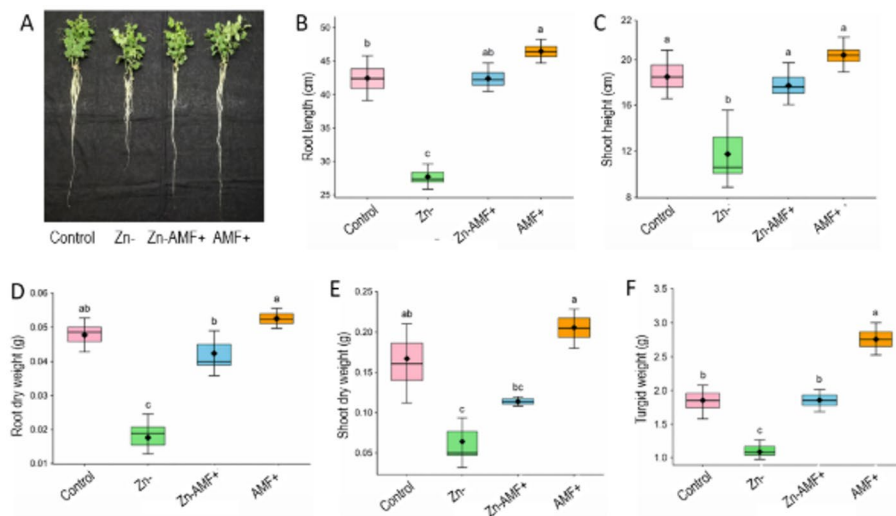
**Fig. 1** Arbuscular mycorrhizal fungal (AMF) colonization, molecular detection, and relative abundance in pea roots under different treatments. **A–D** Fungal colonization in roots stained with trypan blue. **E** PCR amplification of AMF *18S rRNA* gene; M, 100 bp DNA ladder; lane 1, control; lane 2, Zn–; lane 3, Zn–AMF+; lane 4, AMF+. **F** Relative abundance of AMF across treatments. Means  $\pm$  SD (standard deviation) of three independent biological replicates are shown. Different lowercase letters indicate statistically significant differences among treatments ( $P < 0.05$ ), where applicable

observations, the relative abundance of AMF was significantly higher in Zn-AMF+ and AMF+ plants compared with the control and Zn- plants (Fig. 1F). Interestingly, no significant difference in AMF abundance was observed between Zn-AMF+ and AMF+ plants (Fig. 1F).

### 3.2 AMF inoculation improves morphological features under Zn deficiency

Zn deficiency caused visible growth inhibition in plants, characterized by reduced shoot and root development compared to the control plants (Fig. 2A). However, AMF inoculation restored plant growth under Zn-deficient conditions, while AMF+ plants under Zn-sufficient conditions exhibited the greatest overall growth performance (Fig. 2A). Quantitatively, root length significantly declined under Zn deficiency compared to the control treatment (Fig. 2B). In contrast, Zn-deficient plants inoculated with AMF showed a significant improvement in root length relative to Zn– plants. The highest root length was observed in AMF+ plants grown under Zn-sufficient conditions (Fig. 2B). Similarly, shoot height significantly decreased in Zn-deficient plants, whereas AMF inoculation restored shoot height to levels statistically comparable to the control plants (Fig. 2C). AMF+ plants exhibited the highest shoot height among all treatment groups (Fig. 2C).

Root dry weight significantly decreased in response to Zn deficiency compared to the control treatment (Fig. 2D). However, inoculation with AMF under Zn-deficient conditions significantly increased root dry weight relative to Zn-deficient plants without AMF inoculation. Plants supplemented with AMF under Zn-sufficient conditions showed similar root dry weight to those of controls (Fig. 2D). A similar trend was observed for shoot dry weight, where Zn deficiency caused a marked reduction compared to controls, while AMF inoculation showed a non-significant increase in shoot dry weight under Zn-deficient conditions (Fig. 2E). Plants inoculated with AMF without Zn deficiency showed similar shoot dry weight to those of controls (Fig. 2E). Turgid weight



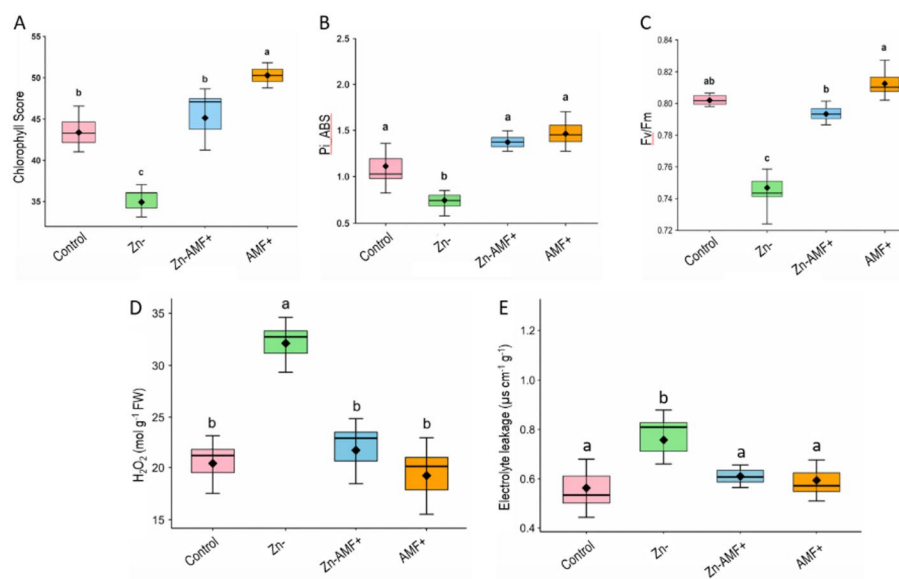
**Fig. 2** Effect of AMF (arbuscular mycorrhizal fungi) inoculation on **A** phenotype of plants, **B** root length, **C** shoot height, **D** root dry weight, **E** shoot dry weight, and **F** shoot turgid weight of plants cultivated under control, Zn-, Zn-AMF+, and AMF+ treatments. Means  $\pm$  SD (standard deviation) of three independent biological replicates are shown. Different lowercase letters indicate statistically significant differences among treatments ( $P < 0.05$ ), where applicable

also significantly declined under Zn deficiency compared to the control plants (Fig. 2F). AMF inoculation significantly increased turgid weight in Zn-deficient plants relative to Zn-deficient plants without AMF. Furthermore, AMF+ plants under Zn-sufficient conditions exhibited the highest turgid weight among all treatment groups (Fig. 2F).

### 3.3 AMF enhances physiological responses under Zn-deficient conditions

Zn deficiency significantly reduced chlorophyll score compared to the control plants (Fig. 3A). However, AMF inoculation under Zn-deficient conditions significantly improved chlorophyll scores relative to Zn-deficient plants without AMF inoculation. Plants grown under AMF+ conditions exhibited the highest chlorophyll score among all treatment groups (Fig. 3A). Furthermore, the photosynthetic performance index significantly declined in Zn-deficient plants compared to controls (Fig. 3B). In contrast, AMF inoculation significantly increased the photosynthetic performance index under Zn-deficient conditions, restoring values comparable to the AMF+ treatment. Similarly, the quantum efficiency of photosystem II significantly decreased in Zn-deficient plants relative to controls, indicating impaired photosynthetic efficiency (Fig. 3C). AMF inoculation under Zn deficiency significantly improved quantum efficiency of photosystem II compared to Zn-deficient plants without AMF inoculation, while AMF+ plants exhibited the highest quantum efficiency of photosystem II values among the treatments (Fig. 3C).

Hydrogen peroxide accumulation significantly increased in Zn-deficient plants compared to the control treatment (Fig. 3D). However, AMF inoculation significantly reduced hydrogen peroxide levels under Zn-deficient conditions relative to Zn-deficient plants without AMF inoculation. Plants grown under AMF+ conditions showed hydrogen peroxide levels comparable to the control plants (Fig. 3D). Electrolyte leakage significantly increased under Zn deficiency compared to controls, indicating enhanced membrane damage (Fig. 3E). In contrast, AMF inoculation significantly reduced



**Fig. 3** Effects of Zn deficiency and AMF (arbuscular mycorrhizal fungi) inoculation on **A** leaf chlorophyll score, **B** leaf photosynthetic performance index (PI\_ABS), **C** maximum quantum efficiency of PSII (Fv/Fm) in leaves, **D** root hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) content, and **E** root electrolyte leakage under control, Zn-, Zn-AMF+, and AMF+ treatments. Means ± SD (standard deviation) of three independent biological replicates are shown. Different lowercase letters indicate significant differences among treatments ( $P < 0.05$ )

electrolyte leakage in Zn-deficient plants relative to Zn-deficient plants without AMF inoculation. AMF+ plants showed electrolyte leakage levels similar to the control treatment (Fig. 3E).

### 3.4 AMF inoculation leads to improvement in nutritional status

Nutrient analysis revealed significant alterations in mineral accumulation in both roots and leaves under Zn deficiency and AMF inoculation conditions (Table 1). In roots, Cu concentration significantly increased in Zn-deficient roots relative to controls, whereas AMF inoculation significantly reduced Cu accumulation compared to Zn-deficient plants without AMF inoculation (Table 1). Mn concentration in roots significantly increased under Zn deficiency compared to the control treatment. AMF inoculation further enhanced Mn accumulation under Zn-deficient conditions, showing the highest Mn concentration among all treatment groups. In contrast, Zn concentration significantly decreased in Zn-deficient roots relative to controls, while AMF inoculation restored Zn levels under Zn-deficient conditions compared to the plants solely grown under Zn deficiency. AMF+ plants showed similar Zn levels to those of controls and Zn-AMF+ plants (Table 1). Root Fe concentration significantly increased under Zn deficiency compared to controls. However, AMF inoculation significantly reduced Fe accumulation in roots under Zn-deficient conditions compared to Zn-deficient plants, with the lowest Fe concentration observed in AMF+ plants (Table 1).

In leaves, Fe concentration significantly increased under Zn deficiency relative to the control plants. AMF inoculation reduced Fe accumulation under Zn-deficient conditions, restoring levels closer to those of controls (Table 1). Leaf Cu concentration also significantly increased under Zn deficiency compared to controls, whereas AMF inoculation partially reduced Cu accumulation. The highest Cu concentration was observed in AMF+ plants under Zn-sufficient conditions (Table 1). Leaf Mn concentration

**Table 1** ICP-MS (Inductively Coupled Plasma Mass Spectrometry) analysis of mineral contents in root and leaf of pea cultivated under different Zn and AMF (arbuscular mycorrhizal fungi) treatments

| Elements (ppm) | Root                     |                          | Leaf                     |                          |                          |                          |
|----------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|                | Control                  | Zn-                      | Zn- AMF+                 | Control                  | Zn-                      | Zn- AMF+                 |
| Cu             | 19.01±1.73 <sup>b</sup>  | 23.60±1.05 <sup>a</sup>  | 16.07±0.33 <sup>c</sup>  | 72.8±0.31 <sup>d</sup>   | 114.3±0.28 <sup>b</sup>  | 98.1±0.21 <sup>c</sup>   |
| Mn             | 28.86±0.39 <sup>d</sup>  | 40.32±0.30 <sup>c</sup>  | 53.59±0.43 <sup>a</sup>  | 375.4±0.44 <sup>ab</sup> | 426.1±57.56 <sup>a</sup> | 236.3±0.42 <sup>c</sup>  |
| Zn             | 20.35±1.10 <sup>a</sup>  | 14.65±1.08 <sup>b</sup>  | 21.50±0.79 <sup>a</sup>  | 130.54±0.32 <sup>a</sup> | 98.42±10.00 <sup>d</sup> | 126.82±1.20 <sup>b</sup> |
| Fe             | 275.87±0.29 <sup>b</sup> | 319.55±0.34 <sup>a</sup> | 249.75±0.13 <sup>c</sup> | 446.4±0.23 <sup>b</sup>  | 495.3±0.35 <sup>a</sup>  | 426.3±0.44 <sup>c</sup>  |

Different letters denote significant differences among control, Zn-, Zn-AMF+, and AMF+ conditions according to Duncan's multiple range test ( $P < 0.05$ ). Data are presented as means ± SD (standard deviation) from three independent biological samples ( $n = 3$ )

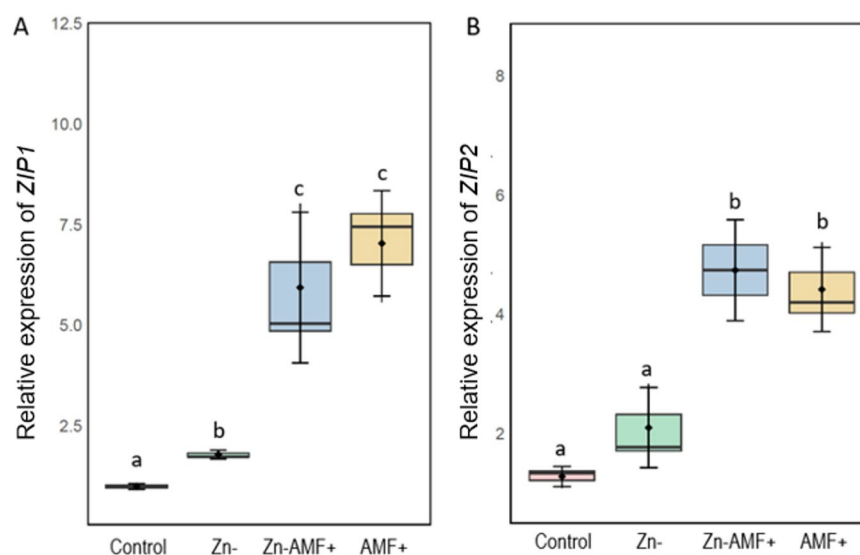
significantly increased under Zn deficiency relative to controls. However, AMF inoculation markedly reduced Mn accumulation under Zn-deficient conditions. In contrast, Zn concentration significantly declined in leaves under Zn deficiency, whereas AMF inoculation significantly restored Zn levels compared to Zn-deficient plants without AMF inoculation. AMF+ plants showed Zn concentrations comparable to the control treatment (Table 1).

### 3.5 AMF colonization in roots increases the expression of Zn transporter genes

The relative expression of *ZIP1* significantly increased under Zn deficiency compared to the control plants (Fig. 4A). AMF inoculation under Zn-deficient conditions further enhanced the expression of this gene relative to Zn-deficient plants without AMF inoculation. The expression level of *ZIP1* in the roots of plants solely inoculated with AMF was similar to that in controls (Fig. 4A). In addition, the expression of *ZIP2* showed a slight non-significant increase under Zn deficiency compared to the control treatment (Fig. 4B). However, AMF inoculation under Zn-deficient conditions significantly upregulated the expression of *ZIP2* relative to Zn-deficient plants without AMF inoculation. AMF+ plants also exhibited significantly elevated expression levels comparable to the Zn-AMF+ treatment (Fig. 4B).

### 3.6 AMF restores amino acid metabolites in Zn-deficient plants

Root amino acid profiling revealed substantial changes in both essential and non-essential amino acids under Zn deficiency and AMF inoculation conditions. Among the essential amino acids, Zn deficiency significantly increased valine, histidine, lysine, and threonine compared to the control treatment, whereas isoleucine, leucine, methionine, and phenylalanine declined under Zn deficiency. AMF inoculation under Zn-deficient conditions significantly restored isoleucine and leucine levels compared to Zn-deficient plants without AMF inoculation. In contrast, valine and histidine remained lower in



**Fig. 4** Expression of zinc transporter genes in response to AMF (arbuscular mycorrhizal fungi) inoculation under Zn deficiency. **A** Relative expression of *ZIP1* and **B** *ZIP2* under control, Zn-, Zn-AMF+, and AMF+ treatments. Means  $\pm$  SD (standard deviation) of three independent biological replicates are shown. Different lowercase letters indicate statistically significant differences among treatments ( $P < 0.05$ ), where applicable

Zn-AMF+ plants compared to Zn-deficient plants (Table 2). Among the non-essential amino acids, Zn deficiency significantly increased aspartic acid, alanine, cystine, proline, serine, and glutamic acid compared to the control plants. However, glycine and tyrosine significantly declined under Zn deficiency. AMF inoculation under Zn-deficient conditions significantly increased glycine and tyrosine accumulation relative to Zn-deficient plants without AMF inoculation (Table 2). In contrast, cystine showed a marked decline under Zn-AMF+ conditions compared to both control and Zn-deficient plants. Aspartic acid, alanine, serine, and glutamic acid also decreased in Zn-AMF+ plants relative to Zn deficiency alone, although their levels generally remained higher than those of the control treatment. Overall, AMF inoculation modulated root amino acid metabolism under Zn deficiency, particularly by restoring amino acids that were reduced during Zn shortage (Table 2).

In leaves, Zn deficiency significantly reduced valine, isoleucine, leucine, methionine, and phenylalanine compared to the control treatment. In contrast, histidine, lysine, threonine, and arginine significantly increased under Zn deficiency. AMF inoculation under Zn-deficient conditions significantly restored the levels of several amino acids, particularly isoleucine, leucine, and arginine, relative to Zn-deficient plants without AMF inoculation (Table 2). Furthermore, Zn deficiency significantly increased aspartic acid and glutamic acid relative to controls, whereas glycine and tyrosine significantly declined. AMF inoculation under Zn-deficient conditions significantly enhanced glycine and tyrosine accumulation compared to Zn-deficient plants without AMF inoculation. In contrast, glutamic acid decreased in Zn-AMF+ plants relative to Zn deficiency alone. Alanine and proline remained relatively unchanged among the treatments, while cystine accumulation increased under AMF+ conditions. Overall, AMF inoculation alleviated several Zn-deficiency-induced alterations in root and leaf amino acid metabolism, particularly by restoring amino acids that were reduced during Zn stress (Table 2).

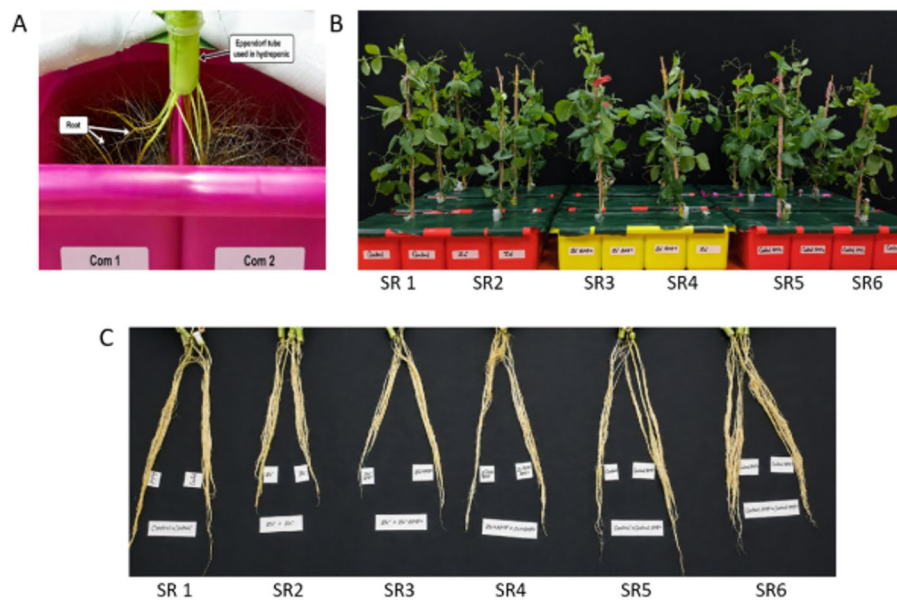
### 3.7 Split-root responses to AMF under Zn deficiency

Split-root treatments caused significant changes in shoot growth and physiological responses under Zn deficiency and AMF inoculation conditions (Fig. 5). Plants grown under Zn-deficient conditions in both compartments (SR2) showed significant reductions in shoot height, shoot dry weight, and chlorophyll score compared to the control treatment (SR1) (Fig. 6A–C). However, AMF inoculation in one or both compartments significantly improved these parameters relative to SR2 plants. Among the treatments, SR6 exhibited shoot height, shoot dry weight, and leaf chlorophyll scores similar to those of the control group (SR1) (Fig. 6A–C). Root length differed between compartments depending on the treatment combinations (Fig. 6D). No significant differences between compartments were observed in SR1, SR2, SR4, and SR6. However, SR3 and SR5 showed significant compartment-specific variation in root length, with the AMF-associated compartments exhibiting greater root elongation compared to the non-inoculated compartments (Fig. 6D). Root dry weight also varied among the split-root treatments (Fig. 6E). Zn-deficient plants (SR2) showed the lowest root dry weight compared to the control treatment. AMF inoculation significantly improved root biomass under Zn-deficient conditions, particularly in SR3 and SR5, where compartment-specific increases were observed. No significant compartmental differences were detected in SR1, SR2, SR4, and SR6 (Fig. 6E). Hydrogen peroxide accumulation showed distinct

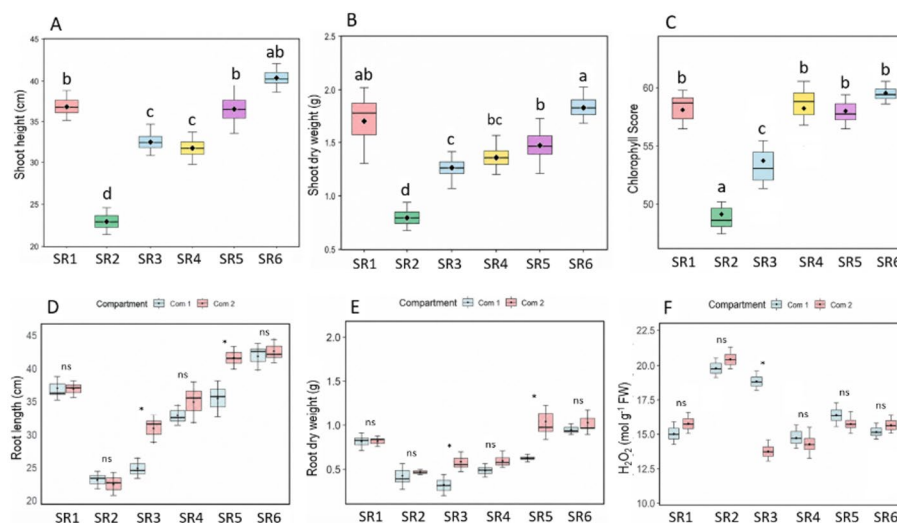
**Table 2** Amino acid analysis in root and leaf of pea plant under different Zn and AMF (arbuscular mycorrhizal fungi) treatments

|                                    | Root                    |                         |                         | Leaf                    |                         |                        |
|------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------------|
|                                    | Control                 | Zn-                     | Zn- AMF+                | Control                 | Zn-                     | Zn- AMF+               |
| Essential amino acids (mg/gDW)     |                         |                         |                         |                         |                         |                        |
| Valine                             | 8.2±0.03 <sup>b</sup>   | 9.3±0.49 <sup>a</sup>   | 7.0±0.61 <sup>c</sup>   | 8.7±0.11 <sup>ab</sup>  | 6.17±0.08 <sup>c</sup>  | 9.11±0.20 <sup>a</sup> |
| Histidine                          | 8.57±0.10 <sup>b</sup>  | 9.91±0.03 <sup>a</sup>  | 8.44±0.68 <sup>b</sup>  | 7.5±0.10 <sup>c</sup>   | 9.02±0.11 <sup>a</sup>  | 9.11±0.10 <sup>a</sup> |
| Isoleucine                         | 5.09±0.10 <sup>a</sup>  | 3.66±0.55 <sup>b</sup>  | 5.47±0.10 <sup>a</sup>  | 6.3±0.08 <sup>b</sup>   | 4.97±0.03 <sup>c</sup>  | 6.92±0.08 <sup>a</sup> |
| Lysine                             | 4.95±0.58 <sup>b</sup>  | 10.68±1.46 <sup>a</sup> | 5.94±0.03 <sup>b</sup>  | 8.6±0.49 <sup>c</sup>   | 11.8±0.97 <sup>a</sup>  | 10.7±0.60 <sup>b</sup> |
| Leucine                            | 10.70±0.05 <sup>b</sup> | 7.28±0.57 <sup>c</sup>  | 11.30±0.03 <sup>a</sup> | 10.38±0.10 <sup>b</sup> | 15.3±1.06 <sup>b</sup>  | 18.9±0.11 <sup>a</sup> |
| Methionine                         | 2.75±0.11 <sup>a</sup>  | 1.59±0.57 <sup>b</sup>  | 2.33±0.10 <sup>ab</sup> | 18.4±0.08 <sup>a</sup>  | 1.85±0.68 <sup>ab</sup> | 2.3±0.10 <sup>ab</sup> |
| Phenylalanine                      | 7.2±0.08 <sup>a</sup>   | 5.56±0.95 <sup>b</sup>  | 7.05±0.08 <sup>a</sup>  | 2.4±0.11 <sup>ab</sup>  | 8.9±0.60 <sup>b</sup>   | 10.3±0.05 <sup>a</sup> |
| Threonine                          | 7.28±0.02 <sup>c</sup>  | 9.2±0.573 <sup>a</sup>  | 7.93±0.02 <sup>b</sup>  | 10.8±0.10 <sup>a</sup>  | 12.7±0.52 <sup>a</sup>  | 11.2±0.03 <sup>b</sup> |
| Arginine                           | 5.89±0.11 <sup>b</sup>  | 5.8±0.10 <sup>b</sup>   | 6.51±0.11 <sup>a</sup>  | 11.3±0.05 <sup>d</sup>  | 12.4±0.03 <sup>b</sup>  | 13.5±0.10 <sup>a</sup> |
| Non-essential amino acids (mg/gDW) |                         |                         |                         |                         |                         |                        |
| Aspartic acid                      | 31.24±0.08 <sup>d</sup> | 34.92±0.08 <sup>a</sup> | 33.57±0.08 <sup>b</sup> | 61.1±0.08 <sup>c</sup>  | 65.3±0.10 <sup>a</sup>  | 63.0±0.10 <sup>b</sup> |
| Alanine                            | 6.70±0.3 <sup>c</sup>   | 7.32±0.03 <sup>a</sup>  | 6.98±0.09 <sup>b</sup>  | 10.4±0.11 <sup>a</sup>  | 10.4±0.10 <sup>a</sup>  | 10.4±0.10 <sup>a</sup> |
| Cystine                            | 26.4±0.10 <sup>b</sup>  | 27.7±0.11 <sup>a</sup>  | 6.6±0.05 <sup>d</sup>   | 33.3±0.03 <sup>c</sup>  | 28.9±0.05 <sup>d</sup>  | 35.0±0.05 <sup>b</sup> |
| Proline                            | 7.47±0.03 <sup>a</sup>  | 7.68±0.91 <sup>a</sup>  | 7.40±0.03 <sup>a</sup>  | 11.0±1.01 <sup>a</sup>  | 11.6±1.00 <sup>a</sup>  | 11.1±0.11 <sup>a</sup> |
| Serine                             | 10.9±0.11 <sup>b</sup>  | 12.1±0.97 <sup>a</sup>  | 10.9±0.10 <sup>b</sup>  | 12.5±0.10 <sup>a</sup>  | 10.7±0.03 <sup>c</sup>  | 11.9±0.10 <sup>b</sup> |
| Glutamic acid                      | 25.4±0.08 <sup>d</sup>  | 28.1±0.10 <sup>a</sup>  | 26.5±0.10 <sup>c</sup>  | 41.5±0.10 <sup>b</sup>  | 43.8±0.03 <sup>a</sup>  | 37.2±0.08 <sup>c</sup> |
| Glycine                            | 6.8±0.10 <sup>b</sup>   | 6.4±0.05 <sup>d</sup>   | 7.04±0.08 <sup>a</sup>  | 12.2±0.03 <sup>a</sup>  | 10.6±0.11 <sup>c</sup>  | 11.9±0.03 <sup>b</sup> |
| Tyrosine                           | 6.6±0.03 <sup>b</sup>   | 4.68±0.63 <sup>d</sup>  | 5.94±0.03 <sup>c</sup>  | 11.7±0.10 <sup>a</sup>  | 7.6±0.10 <sup>d</sup>   | 8.5±0.10 <sup>c</sup>  |

Different letters denote significant differences among control, Zn-, Zn-AMF+, and AMF+ conditions according to Duncan's multiple range test ( $P < 0.05$ ). Data are presented as means ± SD (standard deviation) from three independent biological samples ( $n = 3$ )



**Fig. 5** Experimental setup and phenotypic responses to Zn and AMF (arbuscular mycorrhizal fungi) treatments. **A** Experimental setup: hydroponic split-root system showing root placement in two compartments (Com 1 and Com 2), **B** aboveground growth performance of plants under different Zn and arbuscular mycorrhizal fungi (AMF) treatments and **C** belowground root phenotypic response. Different split-root (SR) treatments are defined as: SR1 = control/control, SR2 = Zn-/Zn-, SR3 = Zn-/ Zn-AMF+, SR4 = Zn-AMF+/Zn-AMF+, SR5 = control/AMF+, and SR6 = AMF+/ AMF+



**Fig. 6** Growth, physiological traits, and compartment-specific responses under different Zn and AMF (arbuscular mycorrhizal fungi) treatments. **A** Shoot height, **B** shoot dry weight, and **C** chlorophyll score across treatments. **D** Root length, **E** root dry weight, and **F** hydrogen peroxide ( $H_2O_2$ ) content measured separately in two compartments (Com 1 and Com 2). Different split-root (SR) treatments are defined as: SR1 = control/control, SR2 = Zn-/Zn-, SR3 = Zn-/ Zn-AMF+, SR4 = Zn-AMF+/Zn-AMF+, SR5 = control/AMF+, and SR6 = AMF+/ AMF+. Means  $\pm$  SD (standard deviation) of three independent biological replicates are shown. Different lowercase letters indicate statistically significant differences among treatments ( $P < 0.05$ ). Asterisks (\*) indicate significant differences between compartments within the same treatment, while "ns" denotes non-significant differences

responses among the split-root treatments (Fig. 6F). Zn deficiency increased hydrogen peroxide levels compared to the control treatment. AMF inoculation reduced hydrogen peroxide accumulation under Zn-deficient conditions, particularly in SR3, where significant compartment-specific differences were detected. In contrast, SR1, SR2, SR4, SR5, and SR6 showed no significant differences between compartments (Fig. 6F).

#### 4 Discussion

Zn deficiency severely impaired plant growth and physiology, as evidenced by reduced growth and photosynthetic parameters in pea plants. These findings are consistent with the well-established role of Zn as an essential micronutrient involved in chlorophyll biosynthesis, photosynthetic electron transport, membrane stability, and enzyme activation [3, 7]. Zn deficiency also increased hydrogen peroxide accumulation and electrolyte leakage, indicating enhanced oxidative stress and membrane injury. Previous studies have shown that Zn deficiency disrupts redox homeostasis and promotes reactive oxygen species (ROS) accumulation, ultimately leading to cellular damage and growth inhibition [8].

AMF inoculation markedly alleviated these detrimental effects by improving growth performance, chlorophyll content, photosynthetic efficiency, and biomass accumulation under Zn-deficient conditions. The improvement in photosynthetic parameters suggests that AMF protected the photosynthetic apparatus from Zn deficiency-induced photoinhibition. AMF are known to improve nutrient acquisition through extensive extraradical hyphal networks that increase the effective absorptive interface and facilitate micronutrient uptake beyond the rhizosphere depletion zone [1, 29]. In addition, AMF can enhance antioxidant defense systems and stabilize cellular membranes under nutrient stress conditions [6]. The reduced hydrogen peroxide and electrolyte leakage observed in AMF-inoculated plants further support the role of AMF in mitigating oxidative damage under Zn stress.

Zn deficiency significantly altered mineral homeostasis in both roots and leaves, particularly affecting Zn, Fe, Cu, and Mn accumulation. The decline in Zn concentration under Zn deficiency confirms impaired micronutrient availability and uptake. Interestingly, AMF inoculation restored Zn levels in both roots and leaves, demonstrating the ability of AMF to facilitate Zn acquisition. A few studies have reported that AMF improve Zn uptake by increasing absorptive surface area and mobilizing sparingly available Zn in soil through hyphal transport [4, 11]. Although other mineral elements were also quantified (data not presented), their levels did not differ significantly among treatments, indicating that the effects of AMF and Zn supplementation on plant mineral nutrition were element-specific rather than reflecting a general alteration of ion homeostasis. Recent studies demonstrated that AMF-mediated stress alleviation was accompanied by improved nutrient acquisition and ion homeostasis, including increased P, K, and Fe accumulation under cold stress [33]. Thus, although other measured elements remained unchanged in the present study, they may still contribute to plant physiological functions under our experimental conditions.

The increased expression of Zn transporter genes (*ZIP1* and *ZIP2*) under AMF inoculation further suggests that AMF-mediated Zn acquisition may involve molecular regulation of Zn transport systems. Zn transporter proteins, particularly ZIP-family transporters, play critical roles in Zn uptake and redistribution during micronutrient

deficiency [16]. The strong induction of these transporter genes under Zn-AMF+ conditions indicates that AMF not only improve Zn availability in the rhizosphere but may also activate host Zn transport pathways to enhance Zn uptake efficiency. Interestingly, Zn deficiency increased Fe and Cu accumulation, while AMF inoculation partially normalized their levels. This phenomenon may reflect the competitive and interconnected relationships among micronutrient transport systems. Excessive Fe accumulation during Zn deficiency has been previously reported and may occur due to non-specific metal transporter activity and disrupted ion homeostasis [28]. Thus, it appears that AMF-mediated restoration of nutrient balance likely contributed to improved physiological stability under Zn stress in pea plants.

Zn deficiency caused substantial changes in amino acid metabolism, particularly increasing stress-associated amino acids such as histidine, lysine, threonine, glutamic acid, and aspartic acid, while reducing several growth-related amino acids including isoleucine, leucine, glycine, and tyrosine. Amino acid reprogramming is a common response to nutrient stress and often reflects metabolic adjustments associated with osmoprotection, nitrogen remobilization, and stress signaling [19]. The accumulation of glutamic acid and aspartic acid under Zn deficiency may indicate enhanced nitrogen redistribution and stress-associated metabolic activity. AMF inoculation restored several amino acids that declined during Zn deficiency, particularly leucine, isoleucine, glycine, and tyrosine. Branched-chain amino acids such as leucine and isoleucine are associated with energy metabolism and stress recovery, while glycine and tyrosine contribute to antioxidative metabolism and secondary metabolite biosynthesis [14]. These findings suggest that AMF modulate metabolic pathways to support cellular recovery and growth under Zn deficiency.

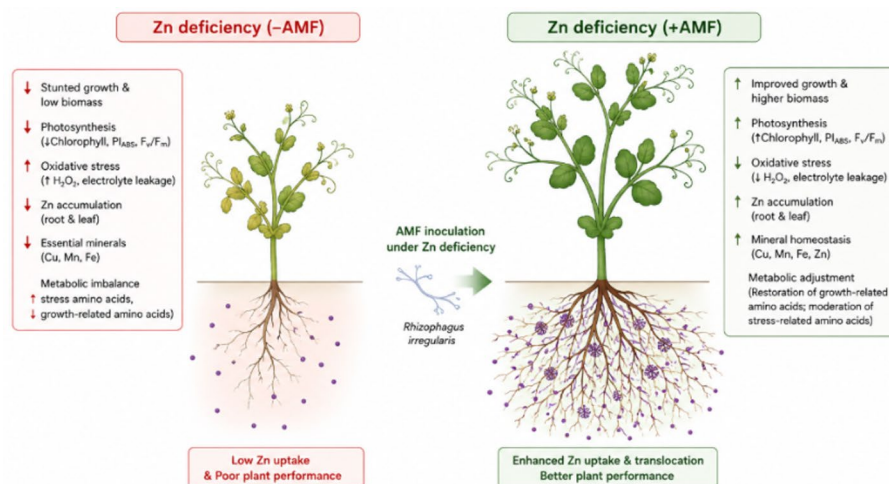
The split-root experiment strongly suggests that AMF-mediated Zn deficiency tolerance operates predominantly through systemic regulation rather than solely through localized root colonization. Although AMF-associated compartments exhibited greater root growth and reduced hydrogen peroxide accumulation, the most striking observation was that improvements in shoot height, shoot dry weight, and chlorophyll score occurred even when only one portion of the root system was inoculated with AMF. This indicates that the beneficial effects of AMF were transmitted beyond the colonized root sector and influenced whole-plant physiology through long-distance signaling pathways. The recovery of shoot growth and chlorophyll status under partial AMF colonization suggests that AMF-induced signaling can systemically regulate photosynthetic performance and biomass accumulation under Zn deficiency. Similar systemic responses have been observed in AMF-associated plants exposed to nutrient and abiotic stresses, where colonized roots trigger whole-plant adaptive mechanisms involving phytohormones, ROS signaling, calcium signaling, and nutrient-responsive pathways [15, 17].

In our study, compartment-specific differences in root traits were more limited compared to the strong shoot-level responses. While some localized increases in root length and root dry weight were detected in AMF-associated compartments, these differences were relatively modest compared to the substantial improvements observed in above-ground traits. Likewise, hydrogen peroxide accumulation was reduced in AMF-colonized compartments, but the reduction in oxidative stress at the whole-plant level suggests systemic regulation of antioxidant responses rather than exclusively local detoxification. These findings imply that direct fungal-root interaction alone cannot fully explain the

observed stress alleviation. Instead, AMF likely induced systemic signaling cascades that coordinated physiological adaptation throughout the plant. In addition, AMF can induce systemic antioxidant responses by modulating ROS-scavenging enzymes and redox signaling pathways, thereby improving membrane stability and reducing oxidative damage under micronutrient stress conditions [6]. The improved shoot performance observed in this study may therefore result from coordinated systemic regulation of nutrient transport, carbon assimilation, and oxidative stress tolerance. Thus, the present study provides evidence that AMF-mediated Zn deficiency tolerance extends beyond localized nutrient acquisition and involves integrated systemic physiological regulation throughout the plant. Despite these findings, several limitations should be considered. All experiments were conducted using pea seedlings under controlled hydroponic conditions, which enabled precise manipulation of Zn availability and AMF inoculation but may not fully represent the complexity of soil-grown plants under field conditions. Moreover, as pea is a legume, nutrient acquisition and stress responses may vary across developmental stages, particularly following the establishment of rhizobial nodulation and symbiotic N fixation, which can substantially alter plant nutrient demand and root physiology. Therefore, the responses observed at the seedling stage may not fully reflect AMF-mediated Zn-deficiency tolerance during later vegetative and reproductive stages. Future studies should validate these responses across different developmental stages and under soil and field conditions. In particular, investigating the interactions among AMF colonization, rhizobial nodulation, N fixation, and Zn homeostasis would provide a more comprehensive understanding of AMF-mediated nutrient regulation in pea.

## 5 Conclusions

*R. irregularis*, a known AMF, substantially improved Zn-deficiency tolerance in pea plants through coordinated regulation of nutrient uptake, physiological performance, and metabolic adjustment. AMF inoculation enhanced plant growth, chlorophyll status, photosynthetic efficiency, and Zn accumulation while reducing oxidative damage and membrane instability under Zn-deficient conditions. The induction of Zn transporter-related genes further suggests that AMF promote Zn acquisition and internal nutrient redistribution in stressed plants (Fig. 7). In addition, AMF-mediated modulation of amino acid profiles indicates an important role of metabolic reprogramming in maintaining cellular stability and stress adaptation during Zn deficiency. The split-root experiment demonstrated that these beneficial effects were largely mediated through systemic signaling, where AMF colonization on one side of the root system improved the physiological performance of the entire plant. Collectively, these findings highlight the multifaceted role of AMF in enhancing plant resilience to micronutrient deficiency and provide new insight into the systemic nature of AMF-mediated stress tolerance. The study further highlights the potential of AMF-based bioinoculants as an environmentally sustainable strategy for improving plant performance under Zn deficiency, although their effectiveness should be validated under soil and field conditions.



**Fig. 7** Mechanistic model illustrating AMF-mediated alleviation of Zn deficiency in pea plants. Zn deficiency reduced plant growth, biomass accumulation, chlorophyll status, photosynthetic efficiency ( $P_i_{ABS}$  and  $F_v/F_m$ ), Zn uptake, and mineral homeostasis, while increasing oxidative stress and metabolic imbalance. In contrast, inoculation with arbuscular mycorrhizal fungi (AMF; *Rhizophagus irregularis*) enhanced Zn acquisition and translocation through extensive hyphal networks and root colonization. AMF inoculation improved plant growth, chlorophyll content, and mineral balance, reduced hydrogen peroxide ( $H_2O_2$ ) accumulation and electrolyte leakage, and restored growth-related amino acids under Zn-deficient conditions. Overall, AMF symbiosis improved physiological stability and Zn-deficiency tolerance in pea plants

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#### Author contributions

SAS conducted most of the experiments, analyzed the data, interpreted the results, and prepared the draft manuscript. MNA assisted with plant cultivation and data recording. MRH assisted with data analysis and manuscript preparation. MSR and FM performed ICP-OES and amino acid profiling. TM, MGM and MMR interpreted the results and revised the manuscript. MRK and AHK mentored the work, coordinated the research, and revised the manuscript.

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#### Data availability

All data generated during this study are included in this published article. Additional datasets are available from the corresponding author upon reasonable request.

#### Declarations

##### Ethics approval and consent to participate

The study was conducted in accordance with institutional, national, and international guidelines and regulations for plant research. Consent to participate: Not applicable. This study did not involve human participants or clinical trials.

##### Consent for publication

Not applicable.

##### Competing interests

The authors declare no competing interests.

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